

TCP/IP Control API Deep Research Report for the Eversolo DMP-A8

Executive summary

The “TCP/IP API” exposed by the **Eversolo DMP-A8** is, in practice, a **local-network HTTP control surface over TCP**—most consistently observed on **port 9529** with a base such as `http://<device-ip>:9529/...` rather than any raw TCP socket protocol. Official Eversolo materials show simple, unauthenticated HTTP GET examples (not TLS) and list remote-control key injection commands, but do **not** publish a full DMP-A8-specific protocol spec. ¹

A much more complete, structured “open API” documentation set is available under **Zidoo’s network open API** site (the device family appears to share an API lineage: endpoints such as `/ZidooControlCenter/...` and `/ZidooMusicControl/v2/...` show up in Eversolo’s own developer snippet and in community integrations). This Zidoo doc set covers: model/identity (`getModel`), now-playing state (`getState`), playback controls, power options, input/output enumeration and selection, and a Wake-on-LAN description. ²

Community implementations (notably a Home Assistant ³ custom integration, and an Unfolded Circle Remote integration) materially extend practical coverage with additional working endpoints (e.g., absolute volume set, mute/unmute, brightness controls, VU/spectrum mode controls) and real-world “gotchas,” including content-type quirks, path inconsistencies, screen on/off behavior, and model/firmware feature gating. ⁴

Wake-on-LAN (WOL) is officially supported on the DMP-A8, but official guidance is **operational** (wired Ethernet + same LAN, wait after sending) rather than protocol-specific; Zidoo’s WOL doc explicitly calls out **broadcast port 9517** and use of the wired MAC address, which is a strong hint but should be packet-captured in your environment before you hard-code it. ⁵

Official documentation and protocol baseline

Transport, security posture, and the key implication for developers

Across the official Eversolo “Developer Platform” page and the Zidoo Open API doc, the control protocol is explicitly **HTTP** (not HTTPS), with examples using a plain `http://<ip>:9529/...` base and no authentication material shown (no tokens, no cookies, no signatures). ¹

That combination implies two practical realities:

1. **Anyone on the same L2/L3 network segment can potentially control the device** if they can reach TCP/9529 (unless you add network segmentation or firewalling yourself). ⁶

2. You should treat the API as **LAN-scoped**: Eversolo's pairing/WOL guidance repeatedly asserts "same local area network," consistent with an expectation of trusted-LAN operation. ⁷

Physical context of "LAN only" requirements

Eversolo's DMP-A8 user manual and WOL guide specify that WOL requires **wired Ethernet**, and that the controller (phone/app) and the device must be on the **same LAN**; they also caution you not to resend WOL repeatedly and to expect compatibility variance across network gear. ⁸

Official Eversolo developer snippet: minimal API surface

Eversolo's Developer Platform page provides:

- A `getModel` example on port 9529 under a `/ZidooControlCenter/getModel` path and showing device identity fields (status, model, MACs, firmware, etc.). ⁹
- A "Remote control" pattern using `/ZidooControlCenter/RemoteControl/...` with `sendkey` and `inputtext`, plus a long list of `Key.*` constants (volume, playback, navigation, power, etc.). ⁹

This is official, but it is **not a complete protocol reference** for music playback state, queues, I/O, volume scaling, etc. ⁹

Official "Eversolo TCP" PDF: legacy but still useful for keycodes

A short official PDF titled "Eversolo TCP" (published for DMP-A6 but widely referenced by the community) documents:

- "TCP" as "HTTP," with base URL on port **9529**. ¹⁰
- The `sendkey` endpoint and a command table including: Poweroff/Reboot, media play/pause/next/prev, mute, volume up/down, output selection keys, and screen display keys. ¹¹

A major developer "gotcha" is visible even in this minimal PDF: it presents `/ControlCenter/RemoteControl/sendkey` as the base, but the example uses `/ZidooControlCenter/RemoteControl/sendkey`, implying multiple historical/alias base paths or a documentation inconsistency. ¹²

Zidoo Open API documentation: the most complete primary protocol spec

Zidoo's "network open API" documentation (v1.0.1) is explicit about:

- **Fixed port: 9529**
- **Base address:** `http://IP:9529`
- HTTP GET examples for the endpoints and JSON response structures. ¹³

While branded as Zidoo, this doc is highly relevant because: - Eversolo's official developer snippet uses the same `/ZidooControlCenter/...` naming style. ⁹ - Community code for Eversolo devices calls `/ZidooMusicControl/v2/...` endpoints described in this Zidoo doc set. ¹⁴

Key endpoints and message shapes from Zidoo's documentation that are directly useful:

Device identity

- `GET /ZidooControlCenter/getModel` (no params), returning identity, firmware, and capability flags (e.g., `ableRemoteBoot`, `hasEqSetting`, etc.). ¹⁵

Remote key injection

- `GET /ZidooControlCenter/RemoteControl/sendkey?key=<Key.*>` returning (in the doc) a JSON object with `{"status": 200}`, and listing many `Key.*` names including power modes, navigation, volume, media keys, etc. ¹⁶
- Note the doc shows this as `text/plain` in the response metadata (a content-type quirk that matters for strict JSON clients). ¹⁶

Now-playing / state

- `GET /ZidooMusicControl/v2/getState`, returning a large JSON payload: playback state, position/duration, metadata (including non-ASCII), and a nested `volumeData` structure with max/current volume, mute state, and a displayed dB string. ¹⁷

Power options and power action

- `GET /ZidooMusicControl/v2/getPowerOption` returns supported power operations (tags like `poweroff`, `reboot`, `timeshutdown`, etc.). ¹⁸
- `GET /ZidooMusicControl/v2/setPowerOption?tag=<tag>` executes the operation. ¹⁹

Playback controls

- `GET /ZidooMusicControl/v2/playNext`, `/playLast`, `/playOrPause`, and `/seekTo?time=<ms>` each return a small success payload (`{"status": 200}` in examples). ²⁰

Input/output enumeration & switching

- `GET /ZidooMusicControl/v2/getInputAndOutputList` returns:
- `inputData[]` items (name, tag, icons, sorted index)
- `outputData[]` items (name, tag, enable flag, icons)
- `outputInfo` with detailed format/support info (especially for HDMI-like outputs) ²¹
- `GET /ZidooMusicControl/v2/setInputList?tag=<inputTag>` both sets the input and returns the updated I/O state payload. ²²
- `GET /ZidooMusicControl/v2/setOutInputList?tag=<outputTag>` sets output and returns the updated I/O state payload. ²³

Wake-on-LAN detail (Zidoo doc)

Zidoo's WOL page is unusually specific: it describes WOL as requiring the wired MAC address, broadcasting to `255.255.255.255`, and it lists **broadcast port 9517**. ²⁴

This is not repeated in Eversolo's DMP-A8 WOL guide (which is operational rather than protocol-level), so you should treat 9517 as a strong hint, not a guarantee for the DMP-A8, until confirmed by capture. ²⁵

Firmware-related functional differences that can affect control behavior

Eversolo's DMP-A8 firmware release notes include device-specific items (e.g., "Optimized room correction function," "Optimized IIS output mode settings"), implying that **available inputs/outputs and their behavior can change across firmware**. ²⁶

Separately, Eversolo's DMP-A8 Settings Menu guide documents that some audio behaviors (e.g., "Volume passthrough mode") require minimum firmware versions (e.g., v1.2.70+ for passthrough mode). Even if the API endpoints remain stable, this can change the meaning of "set volume" or the ability to adjust volume at all on certain outputs. ²⁷

Unofficial and community implementations

Home Assistant custom component: a practical endpoint map (aiohttp-centric)

A widely used community reference is the `hchris1/Eversolo` custom integration for GitHub ²⁸ / Home Assistant, which documents supported entities (power on/off, reboot, screen control, brightness, output selection, playback, etc.) and explicitly states it's tested primarily on DMP-A6, but it is still one of the clearest real-world protocol maps. ²⁹

Its API client:

- Uses default port **9529** and a **1-second default update interval**, indicating that frequent polling is feasible on LAN in at least some setups. ³⁰
- Calls Zidoo-documented endpoints such as:
 - `/ZidooMusicControl/v2/getState` (state)
 - `/ZidooMusicControl/v2/getInputAndOutputList`
 - `/ZidooMusicControl/v2/getPowerOption` ³¹
- Also calls additional endpoints that are **not found in the crawlable Zidoo API pages** but are evidently functional in practice:
 - `/ZidooMusicControl/v2/setDevicesVolume?volume=...` (absolute volume set)
 - `/ZidooMusicControl/v2/setMuteVolume?isMute=0|1`
 - `/ZidooMusicControl/v2/changVUDisplay?openType=...`
 - `/SystemSettings/displaySettings/getScreenBrightness` and `setScreenBrightness?index=...`
 - `/SystemSettings/displaySettings/getKnobBrightness` and `setKnobBrightness?index=...`
 - `/SystemSettings/displaySettings/getVUModeList` and `setVUMode?index=...`
 - `/SystemSettings/displaySettings/getSpPlayModeList` and `setSpPlayModeList?index=...` ³²

Two particularly important implementation details from this integration:

- It parses JSON with `response.json(content_type=None)` (aiohttp), a deliberate workaround for inconsistent/missing content-type headers from the device. ³³
- It determines "screen on/off" state via a heuristic: it calls `getPowerOption` and looks for a `tag == "screen"` option whose localized name contains "Screen off" (or translations). This

suggests the API does **not** expose a direct “screen is on” boolean, at least not in the documented endpoints. ³⁴

Unfolded Circle Remote integration: operational guidance + polling model

The `mase1981/uc-intg-eversolo` integration (for Unfolded Circle Remote 2/3) provides additional pragmatic guidance:

- Confirms **HTTP API on port 9529**, “same local network,” and recommends static IP/DHCP reservation; it also describes “periodic polling for state updates.” ³⁵
- Its accompanying discovery analysis report (for DMP-A6 and DMP-A10, firmware v1.5.46) enumerates additional **SystemSettings** endpoints (brightness, display modes) and notes behavioral quirks:
- “Display Off/On Behavior”: “Display Off just dims ... instead of turning screen completely off” and recommends using remote keys `Key.Screen.OFF/ON` instead of brightness APIs.
- Model-specific “Knob brightness ... doesn’t work” on a model without hardware support.
- Output items that exist but are disabled unless hardware is connected (e.g., “USB DAC output ... enable: false”). ³⁶

Although that discovery run did not include the DMP-A8, it is still valuable evidence that (a) API surface expands beyond Zidoo’s published pages and (b) **feature gating is real**, so you should design around capability discovery at runtime. ³⁷

Reported DMP-A8 integration failure: “reachable via HTTP GET but integration says unknown”

A concrete DMP-A8-related failure report exists in the Unfolded Circle integration issue tracker: users report that raw HTTP GET calls work (for power off, etc.), yet the integration setup shows “state unknown” / “not reachable.” This is evidence of practical failure modes likely tied to polling assumptions, endpoint differences, firmware differences, or HTTP parsing differences. ³⁸

An alternate (non-IP) fallback: learned IR codes

The Unfolded Circle repo also includes learned IR codes (CSV) for common keys (power, play/pause, volume, mute, etc.). While not TCP/IP, it represents a practical “workaround path” if IP control is blocked by network segmentation or firmware quirks. ³⁹

Developer gotchas and recommended mitigations

Path inconsistencies and endpoint aliasing

Symptom: Different documents and implementations use **different base prefixes**: - `/ZidooControlCenter/...` (Eversolo developer snippet; Zidoo sendkey doc) ⁴⁰
- `/ControlCenter/...` (Eversolo TCP PDF base; Home Assistant `getModel`) ⁴¹
- The DMP-A6 PDF even mixes both within the same doc (base says `/ControlCenter`, example uses `/ZidooControlCenter`). ¹²

Mitigation pattern: implement a “base-path fallback” strategy:

- For model: try `GET /ZidooControlCenter/getModel`, then `GET /ControlCenter/getModel`.
- For remote keys: try `GET /ZidooControlCenter/RemoteControl/sendkey?...` first (most consistently documented). ⁴²

Unauthenticated, unencrypted HTTP

Risk: Because the API is HTTP on a fixed LAN port with no described authentication, it is susceptible to on-path sniffing and unauthorized control by any host with network reachability. ⁴³

Mitigations: - Restrict TCP/9529 to a trusted VLAN/subnet at the router/firewall. - If you must control across networks, prefer a VPN (WireGuard/Tailscale-style) rather than port-forwarding. - Avoid exposing `9529/tcp` to the public internet.

Content-type / JSON parsing quirks

Evidence: - Zidoo’s `sendkey` doc shows a `text/plain` response type even though the body is shaped like JSON (`{"status": 200}`). ¹⁶

- The Home Assistant client explicitly parses JSON with aiohttp using `content_type=None`, bypassing content-type checks. ⁴⁴

Mitigation: Always parse responses leniently: - Accept JSON with wrong/missing content-type. - Accept empty `{}` bodies for “command” endpoints (many Zidoo endpoints show `{}` as success). ⁴⁵

- Use timeouts and retries for LAN flakiness rather than assuming server misbehavior.

Output “enable” and tag normalization

Evidence: - `getInputAndOutputList` returns `outputData[].enable` and tags such as `SPDIF`, `XLRRCA`, etc. ²¹

- Community code strips `/` from tags and labels, implying some tags can contain `/` in practice (or at least defensively). ⁴⁴

Mitigation: Don’t assume: - `enable` is strictly boolean vs integer (treat truthy/falsey).

- tags are identifier-safe as file keys / entity IDs; normalize them for storage/display but retain the raw tag for API calls.

“Absolute volume” vs “step volume” vs firmware settings

Evidence: - `getState`’s `volumeData` includes `maxVolume` (example: 200) and `currenttVolume`, suggesting a device-native integer scale plus a displayed dB string. ¹⁷

- Community integration uses both: - Step-based volume via `sendkey?key=Key.VolumeUp|Down` and - Absolute volume set via `/ZidooMusicControl/v2/setDevicesVolume?volume=...`. ³³

- Firmware/UI options such as “volume passthrough mode” can make volume non-adjustable for outputs, and these features are firmware-gated (documented for DMP-A8 settings). ²⁷

Mitigation: - Always read `getState.volumeData` first; respect flags like `isVolumeEnable` where available. ¹⁷

- Prefer absolute volume set when you need determinism, but clamp to `[minVolume, maxVolume]` from state and be prepared for certain outputs to reject changes. ⁴⁶

- Treat “output selection” and “volume set” as a coupled system: selecting a different output can change max volume, mute semantics, or whether volume control is enabled. ⁴⁷

Screen control is surprisingly non-trivial

Evidence: - Community discovery reports that “Display Off” via brightness can behave like dimming, and recommends using `Key.Screen.OFF/ON` remote keys instead. ⁴⁸

- Home Assistant implements both brightness-based control (`setScreenBrightness?index=...`) and explicit key-based screen on/off. ³²

Mitigation: implement screen state and control with a two-layer approach: - Use `Key.Screen.OFF/ON` for “hard” screen toggle semantics. - Use brightness endpoints for dimming/UX polish only. - Don’t assume there is a single canonical “screen is on” boolean; you may have to infer it (as Home Assistant does via `getPowerOption`). ³⁴

WOL timing, port uncertainty, and “don’t spam”

Evidence: - DMP-A8 manual and WOL guide: wired Ethernet required; same LAN; wait after sending; compatibility issues possible; don’t send multiple times. ⁴⁹

- Zidoo WOL doc: broadcast `255.255.255.255`, port **9517**, use wired MAC. ⁵⁰

- Generic WOL references: magic packet structure ($6 \times 0xFF + 16 \times \text{MAC}$) and Wireshark has a WOL protocol/dissector/filter. ⁵¹

Mitigation: - Implement WOL as “send once → wait (e.g., 30–90s) → start HTTP probing with backoff.” - Make the UDP port configurable (default candidates: 9517 per Zidoo doc; also common 9/7). - Packet-capture your phone app’s WOL packet to confirm destination port and addressing on your network before you bake assumptions into code. ⁵²

Source comparison

Source	URL	Reliability	Notes
Eversolo “Developer Platform”	https://eversolo.com/Support/developer/	High	Official minimal API intro: <code>getModel</code> example + <code>sendkey</code> key list; shows HTTP on 9529 and <code>/ZidooControlCenter/...</code> naming

Source	URL	Reliability	Notes
"Eversolo TCP" PDF (DMP-A6)	https://music.eversolo.com/dmp/instruction/Eversolo_DMP-A6_TCP_en_v1.0.pdf	High	Official keycode table + base URL hints; contains path inconsistency (/ <code>ControlCenter</code> vs / <code>ZidooControlCenter</code>)
DMP-A8 User Manual (PDF)	https://music.eversolo.com/dmp/instruction/EVERSOLO-DMP-A8-User-Manual-v1.0.pdf	High	Official operational constraints: phone control, WOL requirements & cautions; not a protocol spec
DMP-A8 WOL guide	https://shop.zidoo.tv/a/support/basic-settings/eversolo-dmp-a8-wake-on-lanwol-guide	High	Official: WOL operation flow, "don't resend," same LAN, wired Ethernet
Eversolo pairing guide	https://shop.zidoo.tv/a/support/basic-settings/eversolo-mobile-tablet-control-app-pairing-guide	High	Official: "auto-scan," QR pairing, manual IP fallback; highlights permissions (local network, Bluetooth, camera)
DMP-A8 firmware changelog page	https://www.eversolo.com/Support/downloadList/target/uXoirEESmeVKKmVViAFMcQ%3D%3D.html	High	Official: firmware changes; indicates evolving behavior (IIS output mode settings, room correction)
Zidoo Open API "doc + getModel example"	https://apidoc.zidoo.tv/319438933e0	High	Primary structured protocol reference: fixed port 9529, base <code>http://IP:9529</code> , JSON examples
Zidoo Open API <code>sendkey</code> page	https://apidoc.zidoo.tv/319428206e0	High	Key list + request/response; shows <code>text/plain</code> response type in doc
Zidoo Open API <code>getState</code> page	https://apidoc.zidoo.tv/319551899e0	High	Detailed now-playing payload including <code>volumeData</code>
Zidoo Open API <code>getInputAndOutputList</code> page	https://apidoc.zidoo.tv/319490384e0	High	Full I/O enumeration payload with tags + <code>outputInfo</code>
Zidoo Open API WOL page	https://apidoc.zidoo.tv/319460467e0	Medium-High	WOL broadcast port 9517 and address; likely applicable but should be confirmed on DMP-A8

Source	URL	Reliability	Notes
Home Assistant custom integration (<code>hchris1/Eversolo</code>)	https://github.com/hchris1/Eversolo	Medium-High	Widely used implementation; reveals undocumented endpoints (absolute volume, mute, SystemSettings) and parsing workarounds
Unfolded Circle Remote integration (<code>uc-intg-eversolo</code>)	https://github.com/mase1981/uc-intg-eversolo	Medium	Operational requirements, polling model, model-specific feature matrix
Discovery analysis file (A6/A10)	https://raw.githubusercontent.com/mase1981/uc-intg-eversolo/main/DISCOVERY_ANALYSIS.md	Medium	Concrete endpoint list + observed quirks (screen off vs dim, disabled outputs) and firmware tag
DMP-A8 “state unknown” issue report	https://github.com/mase1981/uc-intg-eversolo/issues/1	Medium	Evidence of DMP-A8 reachability mismatch: HTTP GET works but integration fails
Learned IR codes CSV	https://raw.githubusercontent.com/mase1981/uc-intg-eversolo/main/Custom%20-%20learned%20IR%20codes_Eversolo%20DMP-A6_codeset_2026-02-04.csv	Medium	Non-IP workaround path for key controls

Reported issues with evidence and mitigations

WOL intermittently fails or requires retries

Evidence: Official docs warn about WOL compatibility issues and explicitly recommend waiting and not repeating the command. ⁴⁹

Likely causes: broadcast filtering, VLAN segmentation, Wi-Fi vs Ethernet mismatch, NIC power state, or destination-port differences across sender implementations. ⁶¹

Mitigations:

Implement WOL with: - configurable UDP port (start with 9517 per Zidoo doc; optionally also try 9/7) - exponential backoff HTTP probing after sending - optional directed broadcast (e.g., `192.168.1.255`) instead of `255.255.255.255` if your router blocks limited broadcast - capture WOL packets to confirm (Wireshark `wol` display filter). ⁶²

API client breaks due to wrong/missing content-type

Evidence: Zidoo `sendkey` is documented as returning `text/plain` despite JSON-like bodies, and Home Assistant bypasses aiohttp's content-type checks. ⁶³

Mitigation: Always parse JSON leniently (aiohttp: `resp.json(content_type=None)`), tolerate `{}` and small status bodies, and treat HTTP 200 as primary success. ⁶⁴

“Screen off” is inconsistent: dims vs off

Evidence: Community discovery notes “Display Off just dims ... use remote keys instead,” and Home Assistant directly uses `Key.Screen.OFF/ON`. ⁶⁵

Mitigation: Prefer `sendkey?key=Key.Screen.OFF|ON` for “hard” behavior; keep brightness endpoints for dimming. ⁶⁶

Feature presence is model/firmware dependent

Evidence: Firmware notes mention A8-specific output-mode changes; community discovery shows disabled outputs that depend on hardware connection (USB DAC). ⁶⁷

Mitigation: Use capability discovery at runtime: - call `getModel` and record `firmware` - call `getInputAndOutputList` and drive UI/entities from returned `inputData/outputData` - treat output selection and volume constraints as dynamic per output. ⁶⁸

DMP-A8 integration “reachable via HTTP GET” but ecosystem integration fails (state unknown)

Evidence: A DMP-A8 user reports HTTP GET works for power off, but the Unfolded Circle integration shows “unknown/not reachable.” ⁶⁰

Likely causes (hypotheses): - integration assumes an endpoint path variant not present on that firmware (`/ControlCenter` vs `/ZidooControlCenter`) ⁶⁹

- JSON parsing differences (content-type / encoding) ³³

- polling too aggressive or timing-related during device power states ⁷⁰

Mitigation: In your own implementation, add: - fallback base-path probing at startup - a “health endpoint” (`getModel`) plus “core endpoint” (`getState`) test - configurable poll interval and timeouts - structured error logging and raw-response dumps to diagnose. ⁷¹

Detailed findings: endpoints, ports, formats, and example code

Canonical ports and addressing

- **HTTP control:** TCP 9529, base `http://<ip>:9529` ⁷²

- **WOL (likely):** UDP broadcast, port **9517** (Zidoo doc) but not explicitly confirmed for DMP-A8; must validate by capture. ⁷³

Common request patterns

The API is heavily GET-oriented, including state changes (power, playback, set output). ⁷⁴

Typical call shape:

- GET /ZidooMusicControl/v2/<verb> (music control)
- GET /ZidooControlCenter/RemoteControl/sendkey?key=<Key.*> (remote injection)
- GET /SystemSettings/... (display settings, in community implementations)

Example curl requests

```
# Device identity
curl -sS "http://$IP:9529/ZidooControlCenter/getModel"

# Now playing / state (big JSON)
curl -sS "http://$IP:9529/ZidooMusicControl/v2/getState"

# Send a remote key (e.g., VolumeUp)
curl -sS "http://$IP:9529/ZidooControlCenter/RemoteControl/sendkey?
key=Key.VolumeUp"

# Next track
curl -sS "http://$IP:9529/ZidooMusicControl/v2/playNext"

# Inputs/outputs enumeration
curl -sS "http://$IP:9529/ZidooMusicControl/v2/getInputAndOutputList"

# Switch input by tag (example tag from list: XMOS)
curl -sS "http://$IP:9529/ZidooMusicControl/v2/setInputList?tag=XMOS"

# Switch output by tag (example: XLR)
curl -sS "http://$IP:9529/ZidooMusicControl/v2/setOutInputList?tag=XLR"

# Power off (tag must exist in getPowerOption)
curl -sS "http://$IP:9529/ZidooMusicControl/v2/setPowerOption?tag=poweroff"
```

The endpoint names and example payloads are directly documented in Zidoo's Open API pages. ⁷⁵

Python (requests) example: robust "probe + call" client

```
import json
import time
```

```

from typing import Any, Dict, Optional
import requests

class EversoloClient:
    def __init__(self, host: str, port: int = 9529, timeout_s: float = 5.0):
        self.base = f"http://{host}:{port}"
        self.timeout_s = timeout_s
        # Prefer /ZidooControlCenter but keep fallback aliases
        self.model_paths = [
            "/ZidooControlCenter/getModel",
            "/ControlCenter/getModel",
        ]

    def _get_json(self, path: str, params: Optional[Dict[str, Any]] = None) ->
Any:
        # Cache-bust to avoid weird intermediary caching on some networks
        params = dict(params or {})
        params.setdefault("_t", int(time.time() * 1000))

        url = f"{self.base}{path}"
        r = requests.get(url, params=params, timeout=self.timeout_s)
        r.raise_for_status()

        # Many endpoints return JSON with non-standard content-type; requests
doesn't care.
        # Some endpoints may return empty {} or bytes; handle both.
        if not r.content:
            return None
        try:
            return r.json()
        except Exception:
            # Fall back to text -> json
            return json.loads(r.text)

    def get_model(self) -> Dict[str, Any]:
        last_err = None
        for p in self.model_paths:
            try:
                data = self._get_json(p)
                if isinstance(data, dict) and data.get("status") in (200,
"200", None):
                    return data
            except Exception as e:
                last_err = e
        raise RuntimeError(f"getModel failed on all known paths: {last_err}")

```

```

def get_state(self) -> Dict[str, Any]:
    return self._get_json("/ZidooMusicControl/v2/getState")

def send_key(self, key: str) -> Any:
    return self._get_json("/ZidooControlCenter/RemoteControl/sendkey",
params={"key": key})

def play_pause(self) -> Any:
    return self._get_json("/ZidooMusicControl/v2/playOrPause")

def next_track(self) -> Any:
    return self._get_json("/ZidooMusicControl/v2/playNext")

def set_output(self, tag: str) -> Any:
    # Some implementations also pass index; tag alone is documented.
    return self._get_json("/ZidooMusicControl/v2/setOutInputList",
params={"tag": tag})

if __name__ == "__main__":
    c = EversoloClient("192.168.1.50")
    print(c.get_model())
    print(c.get_state().get("playingMusic", {}))
    c.send_key("Key.VolumeUp")

```

This code is aligned with the documented endpoint families (Zidoo Open API, Eversolo developer snippet) but includes defensive handling for the aliasing observed in official/community docs. 76

Python (aiohttp) example: async polling + tolerant JSON parsing

```

import asyncio
import time
from typing import Any, Dict, Optional

import aiohttp

class AsyncEversoloClient:
    def __init__(self, host: str, port: int = 9529):
        self.base = f"http://{host}:{port}"

    async def _get(self, session: aiohttp.ClientSession, path: str, params:
Optional[Dict[str, Any]] = None,
        parse_json: bool = True) -> Any:
        params = dict(params or {})
        params.setdefault("_t", int(time.time() * 1000))
        url = f"{self.base}{path}"

```

```

        async with session.get(url, params=params,
timeout=aiohttp.ClientTimeout(total=10)) as resp:
            resp.raise_for_status()
            if not parse_json:
                return await resp.read()
            # Critical: ignore content-type to survive incorrect headers
            return await resp.json(content_type=None)

    async def get_state(self, session: aiohttp.ClientSession) -> Dict[str, Any]:
        return await self._get(session, "/ZidooMusicControl/v2/getState")

    async def send_key(self, session: aiohttp.ClientSession, key: str) -> Any:
        return await self._get(session, "/ZidooControlCenter/RemoteControl/
sendkey", params={"key": key})

    async def main():
        client = AsyncEversoloClient("192.168.1.50")
        async with aiohttp.ClientSession() as session:
            # Poll loop (tune interval to your UX needs)
            for _ in range(5):
                st = await client.get_state(session)
                print("state=", st.get("state"), "pos=", st.get("position"),
"title=", (st.get("playingMusic") or {}).get("title"))
                await asyncio.sleep(1.0)

            # Example command
            await client.send_key(session, "Key.VolumeDown")

if __name__ == "__main__":
    asyncio.run(main())

```

The `content_type=None` pattern is directly supported by the Home Assistant integration's client and reflects real-world behavior for these endpoints. ³³

Suggested test plan and troubleshooting steps

A developer-oriented validation sequence

Establish a repeatable test sequence that records both **firmware** and **API outputs** (store JSON snapshots) so you can compare across firmware updates.

1. Network prerequisites

2. Confirm the DMP-A8 is on wired Ethernet if you plan to use WOL. ⁴⁹

3. Assign a static IP or DHCP reservation; Eversolo documents a static IP configuration flow. ⁷⁷

4. Health check

5. `GET /ZidooControlCenter/getModel` (fallback `/ControlCenter/getModel`) and record:

- model name, firmware string, MAC addresses, capability flags. ⁷⁸

6. State read

7. `GET /ZidooMusicControl/v2/getState`; verify:

- metadata encoding (non-ASCII titles)
- position/duration progression
- volume scale and mute state. ¹⁷

8. Remote key injection sanity

9. Send `Key.VolumeUp` (audible change or observe `volumeData.currentVolume` change).

10. Send `Key.Screen.OFF` and then `Key.Screen.ON` (confirm behavior). ⁷⁹

11. I/O enumeration

12. `GET /ZidooMusicControl/v2/getInputAndOutputList`; verify:

- expected input count and tags (A8 has IIS, multiple analog/digital inputs per its product role)
- output enable flags correspond to physical connections. ⁸⁰

13. I/O switching

14. Set input: `setInputList?tag=<tag-from-inputData>`.

15. Set output: `setOutInputList?tag=<tag-from-outputData>`.

16. Re-read I/O list and `getState` to confirm the change is reflected. ⁸¹

17. Playback controls

18. `playOrPause`, `playNext`, `playLast`, and `seekTo?time=<ms>` (if applicable to current source). ⁸²

19. Volume absolute set (unofficial but widely used)

20. If you choose to use `setDevicesVolume` / `setMuteVolume`, treat them as “soft-undocumented”:

- test on your firmware
- clamp ranges based on `getState.volumeData`
- fall back to key-based volume if rejected. ⁸³

Packet capture: exact commands and filters

Because discovery/WOL transport details can vary, capture traffic while pairing and waking.

Capture HTTP control traffic (TCP/9529)

On a Linux workstation on the same LAN:

```
sudo tcpdump -i <iface> -s 0 -w eversolo_9529.pcap "host <DMP_A8_IP> and tcp port 9529"
```

This is a standard tcpdump pattern (capture full packets, write to pcap). ⁸⁴

To view HTTP-like payloads directly in terminal (quick triage), many workflows use `-A` ASCII printing; if you need only a pcap for Wireshark, prefer `-w`. ⁸⁵

Capture WOL and discovery candidates (UDP)

Given uncertainty, capture multiple likely UDP ports during pairing and wake attempts:

```
sudo tcpdump -i <iface> -s 0 -w eversolo_discovery_wol.pcap \  
"host <DMP_A8_IP> or (udp port 9517 or udp port 9 or udp port 7 or udp port 1900 or udp port 5353)"
```

- UDP/9517 is specifically referenced by the Zidoo WOL doc. ²⁴
- UDP/1900 is SSDP (UPnP discovery). ⁸⁶
- UDP/5353 is mDNS (zeroconf). (Not confirmed for Eversolo control discovery; include to observe.)

In Wireshark: - Use display filter `wol` to find magic packets. ⁸⁷

- Use `ssdp` for SSDP. ⁸⁶

- Use `http && tcp.port == 9529` to focus on control. (HTTP dissector applies because it's plain HTTP, not TLS.) ⁸⁸

Validate WOL packet structure

The “magic packet” structure (6×0xFF then 16×MAC) is documented in `wakeonlan` man pages and Wireshark WOL documentation. ⁵¹

Device-side diagnostics: capturing logs for vendor escalation

If you need to escalate to Eversolo support, Eversolo’s official log capture guide instructs installing a “Realtek Debug” tool APK and collecting the `rtk_dump` folder to share with support. ⁸⁹

Mermaid diagrams for discovery and control flows

Discovery and pairing flow (as observed/expected, with unknowns flagged)

```
flowchart TD
    A[Controller app starts] --> B{Auto-scan finds device?}
    B -- Yes --> C[Device selected in app]
    B -- No --> D[User opens Settings > Network > Phone Control on device]
    D --> E[User scans QR code OR manually enters IP]
    E --> C
    C --> F[Controller begins LAN control over HTTP :9529]
    F --> G[Periodic polling: getState, getInputAndOutputList]
```

This flow is consistent with Eversolo's pairing guide ("auto search," QR scan, manual IP entry) but the exact transport used for "auto-scan" is **unspecified in official docs** and should be confirmed via capture (SSDP/mDNS/broadcast probe are candidates). ⁹⁰

Control and state polling sequence

```
sequenceDiagram
    autonumber
    participant Client
    participant DMP as DMP-A8 (HTTP :9529)

    Client->>DMP: GET /ZidooControlCenter/getModel
    DMP-->>Client: 200 + JSON model/firmware/capabilities

    loop Poll (1s-10s depending on UX)
        Client->>DMP: GET /ZidooMusicControl/v2/getState
        DMP-->>Client: 200 + JSON state/volume/track
    end

    Client->>DMP: GET /ZidooControlCenter/RemoteControl/sendkey?key=Key.VolumeUp
    DMP-->>Client: 200 (often minimal body)

    Client->>DMP: GET /ZidooMusicControl/v2/setOutInputList?tag=XLR
    DMP-->>Client: 200 + updated I/O state
```

Endpoints and response structures are taken from the documented Zidoo Open API pages and Eversolo's own developer snippet (same port and naming patterns). ⁹¹

¹ ⁹ ⁴⁰ <https://eversolo.com/Support/developer/>
<https://eversolo.com/Support/developer/>

2 6 13 15 43 68 71 72 75 76 78 91 <https://apidoc.zidoo.tv/319438933e0>
<https://apidoc.zidoo.tv/319438933e0>

3 35 58 <https://github.com/mase1981/uc-intg-eversolo>
<https://github.com/mase1981/uc-intg-eversolo>

4 14 31 32 33 34 44 64 83 https://raw.githubusercontent.com/hchris1/Eversolo/main/custom_components/eversolo/api.py
https://raw.githubusercontent.com/hchris1/Eversolo/main/custom_components/eversolo/api.py

5 8 49 54 <https://music.eversolo.com/dmp/instruction/EVERSOLO-DMP-A8-User-Manual-v1.0.pdf>
<https://music.eversolo.com/dmp/instruction/EVERSOLO-DMP-A8-User-Manual-v1.0.pdf>

7 52 55 61 **Eversolo DMP-A8 Wake-on-LAN(WOL) Guide – eversolo**
<https://shop.zidoo.tv/a/support/basic-settings/eversolo-dmp-a8-wake-on-lanwol-guide>

10 11 12 41 53 69 https://music.eversolo.com/dmp/instruction/Eversolo_DMP-A6_TCP_en_v1.0.pdf
https://music.eversolo.com/dmp/instruction/Eversolo_DMP-A6_TCP_en_v1.0.pdf

16 42 63 79 <https://apidoc.zidoo.tv/319428206e0>
<https://apidoc.zidoo.tv/319428206e0>

17 46 <https://apidoc.zidoo.tv/319551899e0>
<https://apidoc.zidoo.tv/319551899e0>

18 <https://apidoc.zidoo.tv/319478932e0>
<https://apidoc.zidoo.tv/319478932e0>

19 45 74 <https://apidoc.zidoo.tv/319483283e0>
<https://apidoc.zidoo.tv/319483283e0>

20 28 82 <https://apidoc.zidoo.tv/319618644e0>
<https://apidoc.zidoo.tv/319618644e0>

21 47 80 <https://apidoc.zidoo.tv/319490384e0>
<https://apidoc.zidoo.tv/319490384e0>

22 81 <https://apidoc.zidoo.tv/319511079e0>
<https://apidoc.zidoo.tv/319511079e0>

23 <https://apidoc.zidoo.tv/319515898e0>
<https://apidoc.zidoo.tv/319515898e0>

24 25 50 62 73 <https://apidoc.zidoo.tv/319460467e0>
<https://apidoc.zidoo.tv/319460467e0>

26 67 **EverSolo-Service&Support**
<https://www.eversolo.com/Support/downloadList/target/uXoirEESmeVKKmVViAFMcQ%3D%3D.html>

27 <https://shop.zidoo.tv/a/support/basic-settings/dmp-a8-settings-menu-introduction>
<https://shop.zidoo.tv/a/support/basic-settings/dmp-a8-settings-menu-introduction>

29 57 <https://github.com/hchris1/Eversolo>
<https://github.com/hchris1/Eversolo>

30 70 https://raw.githubusercontent.com/hchris1/Eversolo/main/custom_components/eversolo/const.py
https://raw.githubusercontent.com/hchris1/Eversolo/main/custom_components/eversolo/const.py

36 37 48 59 65 66 https://raw.githubusercontent.com/mase1981/uc-intg-eversolo/main/DISCOVERY_ANALYSIS.md

https://raw.githubusercontent.com/mase1981/uc-intg-eversolo/main/DISCOVERY_ANALYSIS.md

38 60 <https://github.com/mase1981/uc-intg-eversolo/issues/1>

<https://github.com/mase1981/uc-intg-eversolo/issues/1>

39 https://raw.githubusercontent.com/mase1981/uc-intg-eversolo/main/Custom%20-%20learned%20IR%20codes_Eversolo%20DMP-A6_codeset_2026-02-04.csv

https://raw.githubusercontent.com/mase1981/uc-intg-eversolo/main/Custom%20-%20learned%20IR%20codes_Eversolo%20DMP-A6_codeset_2026-02-04.csv

51 [wakeonlan\(1\) - Arch manual pages](#)

https://man.archlinux.org/man/extra/wakeonlan/wakeonlan.1.en?utm_source=chatgpt.com

56 90 <https://shop.zidoo.tv/a/support/basic-settings/eversolo-mobile-tablet-control-app-pairing-guide>

<https://shop.zidoo.tv/a/support/basic-settings/eversolo-mobile-tablet-control-app-pairing-guide>

77 [Eversolo Streamers Static IP Configuration Guide](#)

https://www.eversolo.com/Support/support_guide/guide_target/Un8zEIY1uQneq7k9e%5Bld%5D3ulg%3D%3D.html

84 [Using tcpdump on the command line | pfSense Documentation](#)

https://docs.netgate.com/pfsense/en/latest/diagnostics/packetcapture/tcpdump.html?utm_source=chatgpt.com

85 [How to capture HTTP traffic using Wireshark, Fiddler, or tcpdump](#)

https://support.atlassian.com/atlassian-knowledge-base/kb/how-to-capture-http-traffic-using-wireshark-fiddler-or-tcpdump/?utm_source=chatgpt.com

86 [SSDP](#)

https://wiki.wireshark.org/SSDP?utm_source=chatgpt.com

87 [WakeOnLAN \(WOL\)](#)

https://wiki.wireshark.org/WakeOnLAN?utm_source=chatgpt.com

88 [Wireshark shows SSDP protocol for HTTP request, and not ...](#)

https://networkengineering.stackexchange.com/questions/9455/wireshark-shows-ssdp-protocol-for-http-request-and-not-tcp-protocol?utm_source=chatgpt.com

89 [How to Capture Logs on Eversolo Streamers – eversolo](#)

<https://shop.zidoo.tv/a/support/tools/how-to-capture-logs-on-eversolo-streamers>